7.

Method 1. By the given conditions, the laser frequency incident on sodium is

$$\nu = \frac{c}{\lambda} \ . \qquad (1)$$

For sodium moving relative to positive z, at angle θ, at speed u, the received laser frequency is Doppler-shifted to

$$\nu' = \nu \left(1 + \frac{u}{c}\cos\theta \right); \qquad (2)$$

In terms of wavelength,

$$\lambda' = \frac{\lambda}{1 + \frac{u}{c}\cos\theta} \ . \qquad (3)$$

For resonant absorption, $\lambda' = \lambda_0$ is required; that is,

$$\frac{\lambda}{1 + \frac{u}{c}\cos\theta} = \lambda_0 \ . \qquad (4)$$

Solving (4) gives

$$u\cos\theta = c\,\frac{\lambda - \lambda_0}{\lambda_0} \ ; \qquad (5)$$

Substituting the data gives

$$u\cos\theta = 5.85 \times 10^3 \ \mathrm{m \cdot s^{-1}} \ . \qquad (6)$$

From (6), for θ=30°, a sodium atom has speed

$$u_1 = 6.76 \times 10^3 \ \mathrm{m \cdot s^{-1}} \ ;$$

For $\theta = 45°$, the sodium atom has speed

$$u_2 = 8.28 \times 10^3 \ \mathrm{m \cdot s^{-1}} \ .$$

For motion at angles to z in 30° -45°, resonant sodium atoms have speeds in the range

$$6.76 \times 10^3 \ \mathrm{m \cdot s^{-1}} < u < 8.28 \times 10^3 \ \mathrm{m \cdot s^{-1}} \ . \qquad (7)$$

Momentum is conserved in resonant absorption. Let the recoil speed be V; then

$$Mu - \frac{h}{\lambda} e_z = MV \ . \qquad (8)$$

Here e_z is the unit vector along the z axis. From (8) we obtain

$$u - V = \frac{h}{M\lambda} e_z \ . \qquad (9)$$